

SUPPLY CHAIN RISK ANALYSIS & OPTIMIZATION

Comprehensive Backend Analytical & Optimization Report

Prepared For: B.Tech Capstone Project Evaluators

Date: June 24, 2026

System: Analytical Engine (MySQL Data Warehouse & ML Models)

Executive Summary Dashboard

This report highlights the performance and risk profile of the supply chain network. Data was extracted from the DataCo Smart Supply Chain transactional database, cleaned, transformed, and populated into a Star Schema Data Warehouse. Below are the key KPI metrics.

Total Orders Analyzed	13,112	Total Revenue	\$4,210,552,515.17
Total Profit	\$191,853,920.76	Avg Delivery Delay	0.57 Days
High Risk Orders	3328	Low Risk Orders	6179

Supplier Delay Analysis (Top 5 Delayed)

The following suppliers have the highest average delivery delay. Strategic reviews are suggested for those with ratings below 3.5 or delays exceeding 2 days.

Supplier Name	Rating	Average Delay (Days)	Total Shipments
Krishna Industrial Manufacturing	3.20	0.57	18500

Inventory Replenishment Triggers

The following products have dropped below their reorder points. Replenishment orders should be placed immediately with the recommended quantities to prevent stockouts.

Product Name	Warehouse	Current Stock	Reorder Point	Reorder Qty
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